Rowan Brad Quni-Gudzinas — Research & Technology Leader Portfolio

Rowan Brad Quni-Gudzinas

2026-08-05

Rowan Brad Quni-Gudzinas

Research & Technology Leader — Quantum Computing · AI · National-Scale Systems

ORCID 0009-0002-4317-5604 | Zenodo 649+ publications | ResearchGate Profile
LinkedIn Profile | ISNI 0000-0005-2645-6062 | resume v3.10

> I design architectures, build products, and publish research at the intersection of quantum computing, artificial intelligence, and large-scale systems. Over 20 years of delivering national-scale platforms, founding organizations, and publishing the quantitative case that the quantum computing industry's current trajectory is a thermodynamic dead end — along with a viable alternative.

Quick Navigation

Document	Purpose
RESUME.md	Full chronological resume — experience, education, certifications
PORTFOLIO.md	Deep synthesis of research programs, project impact, and methodology
SKILLS-TECHNOLOGY.md	Expanded skills matrix, technology domains, and expertise map

At a Glance

Metric	
Years of experience	20+
Published works	649+ (open-access on Zenodo)
Cumulative views	14,000+ (Zenodo)
Cumulative downloads	169+ (Zenodo)
Patent portfolio	Quantum architectures, neural networks, foundational computing
Federal R&D managed	$1.5M+ (as certified COR)
Data integration scale	50+ heterogeneous sources across 7 domains
Organizations founded	2 (Empowering Change 501(c)(3), QWAV / QNFO Research)

Impact Highlights

🔬 **Published quantitative analysis showing the quantum industry's trajectory is a thermodynamic dead end — and published a viable alternative.** Dilution refrigerators at millikelvin temperatures provide ~50 µW of cooling. Commercial cryocoolers at 4 Kelvin provide ~1 W — a **20,000× difference**. Published a viable roadmap for 4-Kelvin topological quantum processing using commercially available hardware and twisted high-temperature superconductors.

⚛️ **Designed quantum hardware where fault tolerance is a property of the geometry, not a software protocol.** Ultrametric Quantum Computing (UQC) uses a discrete hierarchical geometry where the mathematics itself prevents small errors from accumulating — potentially eliminating the ~1,000:1 physical-to-logical qubit overhead of conventional error correction.

🏙️ **Led the product that scores neighborhoods across America.** Product Manager for the AARP Livability Index: integrated 50+ data sources across 7 domains to score U.S. neighborhoods nationwide. Cited in 20+ academic and policy studies.

🤖 **Founded an AI nonprofit. Built the product. Made national news.** Took Empowering Change from incorporation through public launch. Built an LLM-powered legal navigation platform. Featured in national media for pioneering AI-driven legal democratization.

📊 **Built the quantitative framework that foreshadows computing's next paradigm shift.** Developed JPCUB — Joules per Computational Unit of Benefit — as a leading indicator retrospectively aligned with 6 historical transitions (vacuum tubes through AI accelerators). Published a history-constrained forecast of computing machine evolution 2026–2050 assessing 7 post-silicon candidates with 12 dated, falsifiable predictions. The forecast identifies quantum error correction as the binding constraint — directly motivating QWAV's hardware-level error suppression architecture.

Target Roles

I am seeking a **VP, Director, or Research Partner** role where I can set technical vision, build teams, and turn breakthrough ideas into market reality. See full target-role matrix in RESUME.md.

Sector	Representative Roles
Quantum Computing	VP/Director of Quantum Research, Architecture, Hardware; Chief Quantum Scientist
AI & Machine Learning	VP/Director of AI Research, Applied AI; Chief AI Scientist
Big Tech & Platforms	Director of Research, VP Emerging Technology, Distinguished Scientist
Defense & National Security	Director of Quantum Research, Chief Scientist (clearance-eligible)
Government Labs / FFRDCs	Director of Quantum Research, Program Director, Senior Technical Advisor
Venture Capital	Research Partner, Technical Fellow, Director of Deep Tech Research
Startups & Scale-ups	CTO, VP Engineering, Head of Research, Chief Product Officer

Contact

- **Email:** rowan.quni@outlook.com
- **ORCID:** 0009-0002-4317-5604
- **ResearchGate:** Rowan Quni-Gudzinas
- **Zenodo:** All publications
- **Location:** Open to remote, hybrid, or on-site. Willing to relocate.

ROWAN BRAD QUNI-GUDZINAS

Research & Technology Leader — Quantum Computing · Artificial Intelligence · National-Scale Systems

rowan.quni@outlook.com | **ORCID:** 0009-0002-4317-5604 | **ResearchGate**

PROFESSIONAL SUMMARY

I design architectures, build products, and publish research at the intersection of quantum computing, artificial intelligence, and large-scale systems. Over 20 years, I have: delivered national-scale platforms (AARP Livability Index); founded an AI nonprofit featured in national media (Empowering Change); and published the quantitative case that the quantum computing industry's current trajectory is a thermodynamic dead end. I bring 649+ publications, a patent portfolio, and a rigorous methodology for epistemic hygiene audits and cross-domain innovation — most recently, 9 papers published in a single week (July 2026) spanning quantum foundations, scaffold-invariant analysis, the adelic qubit architecture, computing paradigm forecasting, and information theory. I am seeking a VP, Director, or Research Partner role where I can set technical vision, build teams, and turn breakthrough ideas into market reality.

IMPACT HIGHLIGHTS

Published quantitative analysis showing that the quantum industry's trajectory is a thermodynamic dead end — and published a viable alternative.

Dilution refrigerators at millikelvin temperatures provide ~50 μW of cooling. Commercial cryocoolers at 4 Kelvin provide ~1 W — a **20,000× difference**. Published a viable roadmap for 4-Kelvin topological quantum processing using commercially available hardware and twisted high-temperature superconductors. Patents developed.

Designed quantum hardware where fault tolerance is a property of the geometry, not a software protocol.

Ultrametric Quantum Computing (UQC) uses a discrete hierarchical geometry where the mathematics itself prevents small errors from accumulating — in contrast to standard architectures that can require on the order of **1,000 physical qubits per logical qubit** for software-level error correction. UQC embeds suppression in the hardware, potentially eliminating this overhead. Published across 50+ documents. Patents developed.

Led the product that scores neighborhoods across America.

Product Manager for the AARP Livability Index: integrated **50+ data sources across 7 domains** to score U.S. neighborhoods nationwide. Full product lifecycle ownership across multiple releases. Platform and methodology cited in **20+ academic and policy studies**. Used nationwide to guide municipal planning, grant allocation, and age-friendly policy.

Founded an AI nonprofit. Built the product. Made national news.

Took Empowering Change from incorporation through public launch. Built an LLM-powered legal navigation platform with three integrated modules for self-represented litigants. Featured in national media for pioneering AI-driven legal democratization.

Validated product design firsthand by serving as a pro se litigant in San Francisco tenant rights cases.

Managed $1.5M in federal R&D contracts with full procurement authority.

Certified Contracting Officer's Representative at U.S. Department of Transportation. Full oversight of technical performance, budgets, milestones, and contractor deliverables. Led key components of a national forecasting model that informs federal infrastructure investment. Clearance-eligible.

649+ publications, a patent portfolio, COR certification, and twenty years of sustained output across government, consulting, nonprofit, and independent research.

Open-access body of work spanning quantum architecture, AI/ML systems, signal processing, epistemic hygiene methodology, and scaffold-invariant analysis — freely accessible on Zenodo and ResearchGate. Patent portfolio covering quantum computing architectures, neural network designs, and foundational computing paradigms.

PROFESSIONAL EXPERIENCE

Principal Investigator & Founder

QWAV / QNFO — Independent Research | 2024–Present

Research Output: Published 649+ scholarly works spanning quantum computing architecture, AI/ML systems, signal processing, epistemic hygiene methodology, scaffold-invariant analysis, and cross-domain methodology — all open-access on Zenodo, ResearchGate, and SSRN. Recent output includes 9 papers in a single week (July 2026) spanning quantum foundations, scaffold-invariant analysis, the adelic qubit architecture, computing paradigm forecasting, and information theory.

Quantum Computing: Published the quantitative case against dilution-refrigerated quantum architectures: ~50 μW of cooling at millikelvin versus ~1 W at 4 Kelvin — a 20,000× gap that fundamentally limits scalability. Published a viable 4-Kelvin topological processing roadmap using 45° twisted Bi-2212 superconductors (predicted thermal stability margin $\Gamma \approx 80$). Designed Ultrametric Quantum Computing (UQC), embedding error suppression in the hardware geometry — potentially eliminating the 1,000:1 qubit overhead of conventional error correction. Proposed Quantum Resonance Computing (QRC), a field-based alternative paradigm. Published across 50+ documents. Patents developed.

Artificial Intelligence: Proposed Prime-Attentive Neural Networks (PANNs), a neural architecture incorporating number-theoretic organizational principles. Published the Alpha Pi Project, a 7-chapter monograph adapting cardiac signal processing techniques to quantum state readout. Identified the fundamental signal-processing ceiling imposed by continuous-state architectures — a finding that directly motivated the UQC design.

Intellectual Property & Credentials: Developed patent portfolio covering quantum architectures (UQC, QRC, 4K topological), neural networks (PANNs), and foundational computing paradigms.

Executive Director & Founder

Empowering Change — 501(c)(3) Nonprofit | San Francisco, CA | 2023–2024

- **Founded the organization and led it from incorporation through public launch.** Built an LLM-powered legal navigation platform with three integrated modules: document drafting, procedural guidance, and legal jargon translation for self-represented litigants.
- **Featured in national media** for pioneering AI-driven legal democratization.
- **Validated product design firsthand** by serving as a pro se litigant in San Francisco tenant rights cases, using direct experience to inform platform requirements.

Product Manager & Senior Methods Advisor

AARP Public Policy Institute | Washington, DC | 2016–2021

- **Led the AARP Livability Index** — a national platform integrating 50+ distinct data sources across 7 domains to score U.S. neighborhoods nationwide. Owned the full product lifecycle from strategy through multiple public releases.
- **Generated measurable policy influence.** Platform and methodology cited in 20+ academic and policy studies. Used nationwide to inform municipal planning, community development grant allocation, and age-friendly policy.
- **Co-authored the Livability Index 2018 report** and multiple AARP policy briefs on housing affordability, transportation access, and age-friendly community development.
- **Managed complex multi-stakeholder relationships** spanning federal agencies, state and local governments, academic partners, advocacy organizations, and internal AARP divisions.

Senior Associate, Data Analytics

Deloitte Consulting LLP | Washington, DC | 2015–2016

- **Led data analytics and machine learning engagements** for federal and public-sector clients. Deployed predictive models and data integration strategies addressing complex organizational challenges.
- **Operated within a top-tier consulting environment,** combining technical delivery with client management, executive communication, and business development.

Data Analyst & Research Manager

U.S. Department of Transportation — Federal Highway Administration (FHWA) | Washington, DC | 2011–2015

- **Managed a federal research portfolio exceeding $1.5M** as certified Contracting Officer's Representative. Full authority for technical oversight, budgeting, milestone tracking, and contractor deliverable acceptance.

- **Led key technical components of the national Tour-Based Model System** for long-distance passenger travel forecasting. Model output directly informs federal infrastructure investment decisions.
- **Coordinated interagency modeling standards** across federal, state, and academic partners. Published peer-reviewed research on multimodal travel behavior and national transportation forecasting.

Earlier Professional Roles

Role	Organization	Period
Platform Data Product Manager	iManage	2022–2023
Senior Product Manager	Epsilon (Publicis Groupe)	2021–2022
Senior Data Scientist	The Advisory Board Company	2016
Consultant	IBI Group	2008–2011
Systems Planner	Delaware Valley Regional Planning Commission	2005–2008
GIS Analyst	Cross County Connection TMA	2004–2005

TECHNICAL EXPERTISE

Area	Capabilities
Quantum Computing	Architecture design (UQC, QRC, 4K topological), cryogenic systems analysis, topological quantum computing, hardware-level error suppression, technology roadmapping, quantum–classical interfaces
AI & Machine Learning	Neural architecture design (PANNs), LLM deployment and productization, applied ML/NLP, predictive modeling, signal processing, AI product development
Product & Platform Leadership	Full product lifecycle, product strategy and roadmap, multi-stakeholder management, go-to-market planning, cross-functional team building, user-centered design
Data & Analytics Architecture	Large-scale data integration (50+ heterogeneous sources), national-scale statistical modeling, geospatial analytics (GIS), real-time pipeline design, federal data standards
Research & Innovation Management	Independent research program design, patent strategy, open science dissemination, technology commercialization, federal grant and contract management (COR)
Cross-Domain Methodology	Structural pattern recognition across disparate systems, complex systems modeling, multi-scale architecture, emergence and feedback analysis, formal isomorphism

Area	Capabilities
	identification, epistemic hygiene auditing, scaffold-invariant decomposition

PUBLICATIONS & INTELLECTUAL PROPERTY

- **649+ scholarly works** published open-access on Zenodo, ResearchGate, and SSRN — freely accessible, no paywalls. Works span preprints, technical reports, and books across quantum architecture, AI/ML, signal processing, epistemic hygiene methodology, and scaffold-invariant cross-domain analysis.
- **Patent portfolio** covering quantum computing architectures (UQC, QRC, 4-Kelvin topological processing), neural network designs (Prime-Attentive Neural Networks), and foundational computing paradigms.
- **Representative works:**
  - *Orchestrating the Quantum Future* — 4-Kelvin quantum computing roadmap (v1.0–v1.1)
  - *Ultrametric Quantum Computation* — UQC architecture series (v0.1–v1.0, 50+ documents)
  - *Quantum Resonance Computing* — Field-based quantum paradigm proposal
  - *The Adelic Qubit* — Manufacturable hardware/software architecture specification (v1.0–v1.1)
  - *The Scaffold-Lock Hypothesis* — How notational choices marginalize formal cross-domain systems
  - *The Thermodynamic Imperative* — Quantitative analysis of quantum scaling limits
  - *The Conditional Advantage* — Epistemic hygiene audit of quantum computational advantage claims
  - *The Re-Entry Thesis* — Structural invariance of self-referential paradoxes across disciplines
  - *JPCUB as a Leading Indicator* — Quantitative computing paradigm-shift forecasting aligned with 6 historical transitions (v1.2)
  - *Computing After Silicon* — History-constrained forecast of computing machine evolution 2026–2050 with 12 falsifiable predictions
  - *Adelic Rate-Distortion Theory* — Extends the Adelic Shannon Theory framework; generalizes rate-distortion to the adele ring (v1.1)
  - *QWAV Commercial Platform Architecture v2.3* — Strategic whitepaper with external verification timeline
  - *Adelic Cross-Domain Program* — Bruhat-Tits trees unify RG flow, bosonic QEC, AdS/CFT, Efimov physics (v3.1)
  - *42 Theses on Pattern-Based Reality* — Synthesis monograph (v3.0, July 2026)
  - *Alpha Pi Project* — 7-chapter monograph: cardiac signal processing → quantum architecture
  - *A General Theory of Process* — Cross-domain isomorphism framework
  - *The Information Spectrum* — Book-length treatment of information theory

EDUCATION

- **M.S., City and Regional Studies** — Rutgers University, Edward J. Bloustein School of Planning and Public Policy (2008)
- **B.A., Urban Studies and Community Planning** — Rutgers University–Camden (2004)

CERTIFICATIONS & PROFESSIONAL IDENTIFIERS

- **Project Management Professional (PMP)** — Project Management Institute (2016)
- **Contracting Officer's Representative (COR)** — Defense Acquisition University (2011) · *Clearance-eligible*
- **American Institute of Certified Planners (AICP)** — American Planning Association (2009)
- **ORCID:** 0009-0002-4317-5604 · **ISNI:** 0000 0005 2645 6062

TARGET ROLES

Sector	Representative Roles	Why This Profile Fits
Quantum Computing	VP/Director of Quantum Research, Architecture, Hardware, or Engineering; Chief Quantum Scientist; Distinguished Engineer	Published architectures (UQC, QRC, 4K topological). Quantitative scaling analysis. 50+ technical documents. Patent portfolio
AI & Machine Learning	VP/Director of AI Research, Applied AI, or ML Engineering; Chief AI Scientist	PANNs architecture. Launched LLM-based platform. Data science leadership at national scale. 649+ publications
Big Tech & Platforms	Director of Research, VP Emerging Technology, Distinguished Scientist, Director of Advanced Technology	Cross-domain expertise (quantum + AI + systems). Product leadership at national scale. Patent portfolio
Defense & National Security	Director of Quantum Research, Chief Scientist, VP Advanced Technology, Senior Technical Fellow or Advisor	COR certification (clearance-eligible). Federal agency experience. Quantum architectures. Complex systems modeling
Government Labs & FFRDCs	Director of Quantum Research, Chief Scientist, Program Director, Senior Technical Advisor	COR certification. Federal program management. Quantum computing R&D. Publication record
Venture Capital	Research Partner, Technical Fellow, Entrepreneur in Residence, Director of Deep Tech Research	Technical evaluation (quantum + AI). 649+ publications. Founded two organizations. Product leadership

Sector	Representative Roles	Why This Profile Fits
Consulting	Director/VP of Quantum Practice, VP of AI & Advanced Analytics, Senior Expert/Fellow, Practice Lead for Emerging Technology	Prior Deloitte experience. Client communication. 649+ publications. Cross-domain methodology
Startups & Scale-ups	CTO, Co-Founder/CEO, VP of Engineering, Head of Research, Chief Product Officer	Founded two organizations. Product leadership (AARP, Empowering Change). Patent strategy. Technical vision. CTO-track: steering technical vision, IP creation, product development for deep-tech, quantum software, and advanced analytics startups
Private Research & Institutes	Director of Research, Principal Investigator, Senior Fellow	Led QNFO/QWAV as Principal Investigator — 649+ publications, patent portfolio, cross-domain research program design. Director-track: leading interdisciplinary initiatives bridging theoretical physics, mathematics, and computational sciences
Financial Services	VP of Quantum or Advanced Computing, Director of Advanced Analytics, Head of Computational Research	Quantum architectures for optimization and risk. Data science at scale. Systems modeling
Policy & Think Tanks	Senior Fellow, Director of Technology or Innovation Policy, Research Director	649+ publications. Federal policy experience (AARP, FHWA). Cross-domain expertise. Senior Fellow track: advising government agencies, defense sectors, and think tanks on thermodynamic/physical limits of computing, post-quantum cryptography, AI ethics

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NOTES

- **Name History:** Formerly known professionally as Brad Gudzinas. All research since 2024 published under Rowan Brad Quni-Gudzinas.
- **Location:** Open to remote, hybrid, or on-site. Willing to relocate.
- **Resume DOI (canonical published version):** 10.5281/zenodo.21796102
- **Download the full portfolio (PDF + source files + provenance bundle):** Zenodo v3.10

Research Portfolio — Deep Synthesis

This document provides a detailed, recruiter-friendly synthesis of my research programs, methodologies, and the intellectual architecture connecting my work across quantum computing, artificial intelligence, and cross-domain systems.

Research Architecture

My work operates across three interconnected layers, each building on the one beneath it:

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┌─────────────────────────────────────────────────────────────┐
│  LAYER 3: Cross-Domain Synthesis & Methodology              │
│  (Scaffold-Invariant Analysis, Epistemic Hygiene,           │
│   Structural Pattern Recognition, Consilience Discovery)    │
├─────────────────────────────────────────────────────────────┤
│  LAYER 2: Application Domains                               │
│  (Quantum Computing, AI/ML, Information Theory,             │
│   Legal Tech, Infrastructure Modeling)                      │
├─────────────────────────────────────────────────────────────┤
│  LAYER 1: Foundational Formalisms                           │
│  (p-Adic Analysis, Ultrametric Geometry, Sheaf Theory,      │
│   Algebraic Topology, Category Theory, Measure Theory)      │
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This is not a grab-bag of papers — it is a coherent research program where formal foundations (Layer 1) generate domain applications (Layer 2), and the patterns observed across domains feed back into cross-domain methodology (Layer 3).

Core Research Programs

1. Ultrametric Quantum Computing (UQC) — 50+ Documents

The Problem: Conventional quantum computing architectures face an existential scaling challenge. Superconducting qubits require millikelvin temperatures (~15 mK), supplied by dilution refrigerators providing ~50 μW of cooling. A fault-tolerant logical qubit requires ~1,000 physical qubits. Even modest quantum computers would need megawatts of room-temperature power to sustain microwatts of cooling — a thermodynamic dead end.

The UQC Solution: UQC embeds error suppression in the hardware geometry itself, using ultrametric (hierarchical, tree-structured) spaces where the strong triangle inequality ($d(x,z) \leq \max\{d(x,y), d(y,z)\}$) provides a natural error barrier. In ultrametric space, small perturbations cannot accumulate across hierarchical boundaries — faults remain localized by the geometry rather than propagating. This potentially eliminates the 1,000:1 physical-to-logical qubit overhead entirely.

Key technical contributions: - **Ultrametric encoding:** Qubits are organized in a p-adic hierarchical structure where logical gates operate on clusters rather than individual qubits - **Hardware-level error suppression:** Fault tolerance emerges from the metric properties of the architecture, not from software-layer error correction codes - **4-Kelvin topological variant:** A parallel track using 45° twisted Bi-2212 high-temperature superconductors at commercially accessible 4K temperatures, with predicted thermal stability margin $\Gamma \approx 80$

Zenodo References: Ultrametric Quantum Computation series (v0.1–v1.0), *Orchestrating the Quantum Future* roadmap series

2. The Thermodynamic Imperative — 4-Kelvin Quantum Roadmap

The Argument: I published a quantitative, system-level analysis arguing that the quantum computing industry's reliance on dilution refrigeration faces a thermodynamic scaling barrier. The analysis is simple but inescapable:

Metric	Dilution Refrigerator	Commercial Cryocooler	Ratio
Operating temperature	~15 mK	4 K	267× warmer
Cooling power	~50 μW	~1 W	20,000× more cooling
Room-temp power input	~10 kW	~7 kW	—
Infrastructure	$500K+, custom installation	Off-the-shelf, rack-mountable	—

The Roadmap: Published a technical roadmap for 4-Kelvin quantum processing using: - Twisted high-temperature superconductors (Bi-2212 at 45° twist angle) - Topological qubit encodings with predicted $\Gamma \approx 80$ thermal stability margin - Commercially available cryocooler hardware (no custom dilution units) - Room for 10,000+ physical qubits in a single rack-mountable unit

Zenodo References: *Orchestrating the Quantum Future* v1.0–v1.1, *The Thermodynamic Imperative*

3. The Adelic Qubit — A Manufacturable Quantum Architecture

The Problem: Quantum architectures remain trapped in an "aesthetic formalism" where designs are mathematically elegant but lack a clear path to fabrication. They specify operations on abstract Hilbert spaces without addressing the physical instantiation problem.

The Solution: The Adelic Qubit is a hardware/software architecture specification where mathematical abstractions correspond to physical counterparts. The "adelic" framework unifies information across all completions of the rational numbers (real, p-adic for every prime p) — a mathematical structure that naturally encodes the hierarchical, multi-scale nature of quantum systems.

Key features: - Specification from device physics through application layer - Manufacturable using existing semiconductor fabrication techniques - Software stack designed for compiler optimization, not manually coded gates - Architecture designed with reference to quantum advantage benchmarking frameworks

Zenodo References: *The Adelic Qubit* v1.0–v1.1 DOI series

4. Adelic Rate-Distortion Theory — Extending the Shannon Foundation

The Problem: Shannon's classical rate-distortion theory operates entirely in the real numbers, optimizing for Euclidean fidelity. But information in nature is encoded across all completions of the rational numbers — real and p-adic for every prime p — and a purely real-valued distortion measure misses structure that a p-adic measure would capture.

The Solution: Extends the Adelic Shannon Theory framework, generalizing rate-distortion theory to the adele ring — the simultaneous product of all completions of $\mathbb{Q}$. Defines a p-adic distortion measure $d_p(x,\hat{x}) = p^{-v_p(x-\hat{x})}$ that penalizes information loss in ultrametric spaces differently than Euclidean loss. Proves the adelic Shannon lower bound: no adelic code can achieve below the information-theoretic minimum that respects all completions simultaneously. Establishes the Gaussian entropic number as the hardest source to compress across all completions. Computational verification with rate-distortion curves confirms the theoretical bounds.

Applications: Directly informs quantum state compression protocols (UQC and Adelic Qubit both operate in p-adic/ultrametric spaces), neural network compression (PANNs use number-theoretic attention patterns), and any information-processing system where hierarchical structure matters.

Zenodo References: *Adelic Rate-Distortion Theory* v1.1

5. Prime-Attentive Neural Networks (PANNs)

The Problem: Transformer architectures scale attention quadratically ($O(n^2)$), creating fundamental computational bottlenecks. Biological neural systems achieve similar feats with architectures that look nothing like transformers.

The Solution: PANNs replace dense attention matrices with sparse, number-theoretically structured attention patterns. By organizing neurons along prime-indexed paths, the network achieves: - Near-linear scaling ($O(n \log n)$ in key configurations) - Natural hierarchical information routing - Built-in inductive biases toward modular structure

The architecture draws on p-adic number theory — the same mathematical framework underlying UQC — creating a unified formal foundation across both quantum and classical computation.

Zenodo References: Prime-Attentive Neural Networks series, *Alpha Pi Project* monograph

6. Quantum Resonance Computing (QRC)

A field-based alternative quantum computing paradigm that treats computation as a resonant process in a structured field, rather than as gate operations on discrete qubits. QRC explores whether the gate model itself — treating computation as a sequence of discrete operations — is the wrong abstraction for quantum systems.

7. JPCUB — Leading Indicator of Computing Paradigm Shifts

The Problem: Computing history is a sequence of substrate transitions — vacuum tubes → transistors → integrated circuits → multicore → GPU → TPU/AI accelerators — each triggered when the prior architecture exhausted a fundamental resource constraint. But how do you know when the next transition is imminent, and which candidate paradigm will win?

The JPCUB Solution: JPCUB (Joules per Computational Unit of Benefit) measures the energy cost per unit of useful computation, normalized across architectures. When JPCUB plateaus for an incumbent, a paradigm shift becomes thermodynamically favored — the physics of energy dissipation forces a change regardless of market dynamics or institutional inertia. Retrospectively aligned with all 6 historical transitions with measurable lead times. Prospectively applied to 7 post-silicon candidates — quantum computing, neuromorphic, optical, reversible, probabilistic, cryogenic CMOS, and heterogeneous convergence — with dated, falsifiable predictions registered for each.

Key finding: The shift from vacuum tubes to transistors was not a lucky accident — it followed the same JPCUB plateau pattern that preceded every subsequent transition. This is a falsifiable, quantitative framework for predicting when computing will change and which direction it will take.

Zenodo References: *JPCUB as a Leading Indicator of Computing Paradigm Shifts* v1.2

8. Computing After Silicon — History-Constrained Forecast, 2026–2050

The Problem: Post-silicon computing forecasting is dominated by vendor roadmaps (each company predicting its own technology will win) and academic wishcasting (each lab predicting its own approach will scale). What would a genuinely independent, evidence-grounded forecast look like?

The Solution: Extends the JPCUB framework into a full history-constrained forecast of computing machine evolution from 2026 through 2050. Synthesizes paradigm-shift history through a cross-domain lens — physics (thermodynamic constraints), information theory (rate-distortion bounds), economics (cost-per-compute trends), and biology (evolutionary selection pressures on competing architectures). Assesses the same 7 post-silicon candidates and registers 12 dated, falsifiable predictions for 2030–2040 verification.

Central finding: Heterogeneous convergence — no single paradigm will "win." AI accelerators, quantum co-processors, and probabilistic architectures will co-exist in a tiered compute fabric, with quantum error correction identified as the binding constraint on the quantum component. This converges with QWAV's UQC architecture, which embeds error suppression in hardware geometry rather than relying on software-level error correction that the forecast projects will remain the bottleneck through 2040.

Zenodo References: *Computing After Silicon* v1.0, *JPCUB as a Leading Indicator* v1.2

9. Epistemic Hygiene & Scaffold-Invariant Analysis

A methodological framework for evaluating claims across domains without being trapped by domain-specific notation or social proof. Key contributions:

- **The Scaffold-Lock Hypothesis:** Argues that notational and formal choices (the “scaffolding”) in one domain systematically marginalize structurally identical insights from other domains
- **The Re-Entry Thesis:** Shows that self-referential paradoxes (Liar paradox, Gödel incompleteness, Russell’s paradox, measurement problem) share a common structural invariant — re-entry of a system’s output into its input — and that recognizing this invariant dissolves apparent conflicts
- **The Conditional Advantage:** A systematic epistemic hygiene audit of quantum computational advantage claims, distinguishing genuine physical speedup from aspirational forecasting, benchmark cherry-picking, and classical-algorithm straw-manning
- **42 Theses on Pattern-Based Reality:** A synthesis monograph connecting pattern-first ontology across physics, computation, biology, and social systems

Zenodo References: *The Scaffold-Lock Hypothesis*, *The Re-Entry Thesis*, *The Conditional Advantage*, *42 Theses on Pattern-Based Reality* v3.0

10. Consilience Methodology — Cross-Domain Innovation

A systematic framework for discovering structural isomorphisms across seemingly unrelated domains and translating them into novel research programs. Rather than relying on serendipitous analogies, the consilience methodology:

1. Identifies structural patterns (not surface analogies) that recur across domains
2. Translates concepts across domain lexicons while preserving mathematical structure
3. Generates testable predictions by asking: “If X works in domain A, what should we observe in domain B?”
4. Produces novel research programs that no single domain would have generated on its own

This methodology has produced research programs including: cardiac signal processing → quantum state readout (the Alpha Pi Project), ultrametric geometry → quantum error correction (UQC), and number theory → neural architecture (PANNs).

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11. QWAV Commercial Platform Architecture

The Problem: Quantum computing startups pitch separate hardware and software stacks with no unified commercial architecture showing how research outputs become product capabilities on a defined, externally verifiable timeline.

The Solution: A strategic architecture whitepaper defining QWAV’s full technology stack — from device physics through application layer — with a product roadmap anchored to independently verifiable milestones. The July 2026 v2.3 update cross-references the *Computing After Silicon* forecast (DOI 10.5281/zenodo.21713202), establishing the external falsification timeline against which QWAV’s technology milestones can be independently tracked. The forecast identifies quantum error correction as the binding constraint on the post-silicon trajectory through 2040 — and QWAV’s UQC architecture addresses this directly through hardware-level error suppression, bypassing the software-layer bottleneck entirely.

Key features: Complete technology stack definition, independent external verification timeline via falsifiable forecast predictions, manufacturing-pathway specifications for 4-Kelvin topological processing, patent portfolio alignment.

Zenodo References: *QWAV Commercial Platform Architecture* v2.3

12. Adelic Cross-Domain Program — Bruhat-Tits Trees and the Standard Model

The Problem: The renormalization group, quantum error correction, holographic AdS/CFT duality, Efimov physics, and the Standard Model mass spectrum appear to be unrelated phenomena — different theories in different domains using different mathematics. But what if they share a single geometric substrate?

The Solution: A six-avenue unified synthesis revealing that these phenomena share a common geometric structure: the Bruhat-Tits tree of p-adic numbers — a regular infinite tree encoding the ultrametric completions of the rational numbers. The Bruhat-Tits tree is not a metaphor — it is the actual mathematical object on which RG flow lines are geodesics, bosonic QEC codes (cat, GKP, binomial) are fixed points, and AdS/CFT is a special case of p-adic holography.

Key results: - **Bosonic QEC codes as tree fixed points:** Cat, GKP, and binomial codes correspond to vertices at specific depths on the Bruhat-Tits tree — error correction performance is determined by tree level. - **The Efimov effect is the three-body manifestation of adelic structure:** The infinite tower of three-body bound states emerges from the same p-adic hierarchical structure that organizes QEC codes. - **π is NOT an idèle:** The transcendental field-crossing character of π is a structural necessity — it cannot be embedded in the adele ring as an idèle, which is why RG flows cross completion boundaries rather than staying within a single p-adic field.

v3.1 published July 2026; v3.3 corrections complete (August 2026).

Zenodo References: *Adelic Cross-Domain Program* v3.1 (DOI 10.5281/zenodo.21539547)

Product & Platform Leadership

AARP Livability Index

The Livability Index is a national-scale platform that scores neighborhoods and communities across the United States across seven categories of livability: housing, neighborhood, transportation, environment, health, engagement, and opportunity.

My role: Product Manager and Senior Methods Advisor. Full product lifecycle ownership across multiple public releases.

Technical scale: - 50+ distinct data sources integrated into a unified scoring framework - National coverage at the census tract and block-group level - Customizable weighting engine allowing users to adjust category importance - Public-facing web platform with API access for researchers

Impact: Cited in 20+ academic and policy studies. Used by municipal planners, community development organizations, and grant-makers nationwide. Informed the World Health Organization's Age-Friendly Cities framework adaptation for the United States.

Empowering Change — AI for Legal Access

Founded and led a 501(c)(3) nonprofit building an LLM-powered legal navigation platform for self-represented litigants. The platform provided:

1. **Document Drafting:** AI-assisted generation of court filings, motions, and responses
2. **Procedural Guidance:** Step-by-step navigation of legal processes in plain English
3. **Legal Jargon Translation:** Real-time translation of legal terminology into accessible language

Validation: Product design was directly informed by my experience as a pro se litigant in San Francisco tenant rights cases — using the product myself to validate its utility.

Media: Featured in national media coverage of AI-driven legal democratization.

FHWA National Travel Forecasting Model

Led key technical components of the national Tour-Based Model System, a large-scale agent-based simulation of long-distance passenger travel in the United States. Managed $1.5M+ in federal R&D contracts with full procurement authority as certified Contracting Officer's Representative.

Publication Metrics

By Domain

Domain	Approximate Publications	Key Outputs
Quantum Computing	~190	UQC, QRC, Adelic Qubit, 4K Roadmap, Thermodynamic Imperative
AI & Machine Learning	~120	PANNs, Alpha Pi Project, LLM systems
Cross-Domain Methodology	~150	Scaffold-Lock, Re-Entry, Consilience, Epistemic Hygiene
Information Theory & Signal Processing	~100	Adelic Rate-Distortion, Information Spectrum

Domain	Approximate Publications	Key Outputs
Systems & Infrastructure	~60	Transportation modeling, data integration, GIS
Synthesis & Meta-Work	~40	42 Theses, General Theory of Process, portfolio syntheses

By Type

Type	Count
Technical Reports & Working Papers	~430
Preprints	~100
Monographs & Books	~40
Meta-Analyses & Syntheses	~30
Datasets & Software	~50

Zenodo Statistics (All Publications)

- **Total publications:** 649+ (live Zenodo search count as of July 2026)
- **Cumulative views:** 14,000+
- **Cumulative downloads:** 169+
- **Most viewed paper:** Resume (14,000+ views, accumulated across all versions)

What Makes This Profile Distinctive

1. Rigor Backed by Volume

Many researchers claim cross-domain expertise. I have published 649+ documents across quantum computing, AI, information theory, and systems engineering — the paper trail is public, peer-accessible, and internally consistent. A recruiter or technical interviewer can trace the intellectual architecture directly through the publications.

2. Theory-to-Product Pipeline

I do not just publish papers — I build products that operate at national scale. The AARP Livability Index, Empowering Change platform, and FHWA forecasting model are not proofs of concept; they are deployed systems used by millions of people and cited in policy and academic literature. The quantum architectures (UQC, Adelic Qubit) are designed with manufacturing pathways, not just mathematical elegance.

3. Thermodynamic Grounding

My quantum computing work is distinguished by its grounding in engineering reality. The thermodynamic scaling argument is not a theoretical critique — it is a quantitative analysis using published specifications for commercially available hardware. This is the kind of systems-level thinking that distinguishes a research leader from a theorist.

4. Epistemic Rigor

The epistemic hygiene methodology — scaffold-invariant analysis, structural pattern recognition, consilience discovery — provides a repeatable framework for evaluating claims across domains. This is directly applicable to: technical due diligence (VC), technology roadmapping (corporate R&D), research program design (government labs), and competitive intelligence (strategy).

5. Open Science by Default

All 649+ publications are open-access with permanent DOIs. No paywalls. No institutional access requirements. This is not just a philosophical commitment — it means my entire body of work is immediately verifiable by any hiring manager, technical interviewer, or due diligence analyst.

Selected Publication DOIs

Title	DOI	Domain
Orchestrating the Quantum Future v1.0	10.5281/zenodo.21016993	Quantum
The Adelic Qubit v1.1	10.5281/zenodo.21221823	Quantum
The Scaffold-Lock Hypothesis	10.5281/zenodo.21282108	Methodology
The Conditional Advantage	10.5281/zenodo.21304444	Methodology
The Re-Entry Thesis	10.5281/zenodo.21254993	Methodology
42 Theses on Pattern-Based Reality v3.0	10.5281/zenodo.21389470	Synthesis
JPCUB as a Leading Indicator	10.5281/zenodo.21716180	Computing
Computing After Silicon	10.5281/zenodo.21713202	Computing
Adelic Rate-Distortion Theory	10.5281/zenodo.21710936	Information Theory
QWAV Commercial Platform Architecture v2.3	10.5281/zenodo.21713222	Systems
Adelic Cross-Domain Program v3.1	10.5281/zenodo.21539547	Cross-Domain

All DOIs resolve to open-access publications on Zenodo. For the complete publication list, visit Zenodo or ResearchGate.

This portfolio synthesizes work published under Rowan Brad Quni-Gudzinas (2024–present) and earlier professional work. All research publications are open-access on Zenodo.

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Skills & Technology — Expanded Matrix

This document maps my expertise across technology domains with concrete evidence — publications, products, or certifications — for each claim.

Quantum Computing

Capability	Depth	Evidence
Quantum architecture design	Expert	Three published architectures: UQC (50+ docs), QRC, Adelic Qubit (v1.0–v1.1)
Cryogenic systems analysis	Expert	Quantitative thermodynamic analysis of dilution vs. cryocooler regimes — 20,000× cooling power differential documented
Topological quantum computing	Advanced	4-Kelvin topological roadmap: 45° twisted Bi-2212, Γ ≈ 80 thermal stability prediction
Hardware-level error suppression	Expert	UQC embeds fault tolerance in ultrametric geometry — potentially eliminates 1,000:1 overhead
Quantum error correction	Advanced	Analysis of surface codes vs. geometric suppression; ultrametric encoding theory
Technology roadmapping	Expert	Published product roadmaps: *Orchestrating the Quantum Future*, *QWAV Commercial Platform Architecture v2.3*

Capability	Depth	Evidence
Quantum–classical interfaces	Advanced	Adelic framework for unifying quantum and classical computation
Quantum advantage evaluation	Expert	*The Conditional Advantage* — systematic epistemic hygiene audit of computational advantage claims
Patent strategy	Advanced	Patent portfolio: UQC, QRC, 4K topological, PANNs, foundational computing

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Artificial Intelligence & Machine Learning

Capability	Depth	Evidence
Neural architecture design	Advanced	Designed Prime-Attentive Neural Networks (PANNs) — number-theoretic attention patterns for near-linear scaling
LLM deployment & productization	Advanced	Built and launched LLM-powered legal navigation platform (Empowering Change)
Applied ML/NLP	Advanced	Legal document drafting, procedural guidance, jargon translation modules
Predictive modeling	Expert	National-scale travel demand forecasting (FHWA), Deloitte federal analytics engagements
Signal processing	Advanced	*Alpha Pi Project* — 7-chapter monograph: cardiac signal processing → quantum state readout
AI product development	Expert	Full product lifecycle: Empowering Change platform, AARP Livability Index analytics
Training & fine-tuning	Proficient	LLM configuration and deployment for production systems

Capability	Depth	Evidence
MLOps	Proficient	Data pipeline design at national scale (50+ heterogeneous sources)

Product & Platform Leadership

Capability	Depth	Evidence
Full product lifecycle	Expert	AARP Livability Index (multiple releases), Empowering Change (0→launch), iManage platform
Product strategy & roadmap	Expert	QWAV commercial platform architecture, quantum computing roadmaps
Multi-stakeholder management	Expert	Coordinated federal agencies, states, localities, academia, advocacy orgs (AARP, FHWA)
Go-to-market planning	Advanced	Empowering Change: nonprofit launch strategy + national media coverage
Cross-functional team building	Advanced	Led teams across data science, engineering, policy, and stakeholder functions
User-centered design	Expert	Pro se litigation → product design feedback loop (Empowering Change)
Technical writing for executives	Expert	649+ publications; policy briefs; technical roadmaps; architecture specifications

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Data & Analytics Architecture

Capability	Depth	Evidence
Large-scale data integration	Expert	Integrated 50+ heterogeneous data sources across 7 domains (AARP Livability Index)
National-scale statistical modeling	Expert	Tour-Based Model System (FHWA) — agent-based

Capability	Depth	Evidence
		simulation of U.S. long-distance travel
Geospatial analytics (GIS)	Expert	GIS Analyst (2004–2005) → national-scale spatial data integration (AARP, FHWA)
Real-time pipeline design	Advanced	Platform data product management (iManage, Epsilon)
Federal data standards	Expert	Interagency modeling standards coordination (FHWA), AICP certification
Data visualization	Advanced	Public-facing dashboard design (Livability Index), policy brief preparation

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Research & Innovation Management

Capability	Depth	Evidence
Independent research program design	Expert	Designed and executed 7 coherent, interconnected research programs over 2 years
Patent strategy	Advanced	Patent portfolio: quantum architectures, neural networks, foundational computing
Open science dissemination	Expert	649+ open-access publications with permanent DOIs on Zenodo
Technology commercialization	Advanced	Product roadmaps with manufacturing pathways (Adelic Qubit, 4K topological)
Federal grant & contract management	Expert	COR-certified; managed $1.5M+ in federal R&D contracts with full procurement authority
Technical due diligence	Expert	Epistemic hygiene methodology directly applicable to VC/diligence evaluation
Team building & mentoring	Advanced	Led research teams across QWAV, Empowering Change, and prior roles

Cross-Domain Methodology

Capability	Depth	Evidence
Structural pattern recognition	Expert	Identified ~20 cross-domain structural isomorphisms producing novel research programs
Complex systems modeling	Expert	Multi-scale architecture: ultrametric spaces → quantum computing + neural networks
Emergence & feedback analysis	Expert	*The Re-Entry Thesis* — structural invariant across self-referential paradoxes
Formal isomorphism identification	Expert	Cardiac signal processing ↔ quantum state readout; number theory ↔ neural architecture
Epistemic hygiene auditing	Expert	Published methodology: *The Conditional Advantage*, Scaffold-Lock Hypothesis
Scaffold-invariant decomposition	Expert	*The Scaffold-Lock Hypothesis* — systematic method for identifying notation-induced blind spots
Consilience discovery	Expert	Framework for generating cross-domain research programs from structural isomorphisms
Computing paradigm forecasting	Expert	JPCUB framework retrospectively aligned with 6 historical transitions; 12 prospective predictions registered
History-constrained forecasting	Expert	*Computing After Silicon* — cross-domain synthesis (physics + information theory + economics + biology)

Programming & Tools

Category	Technologies
Languages	Python (primary), JavaScript/TypeScript, SQL, R, MATLAB, Bash/PowerShell
Data Science	pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Jupyter
GIS & Spatial	ArcGIS, QGIS, PostGIS, spatial statistics
Cloud & Infrastructure	AWS, Cloudflare (Workers, D1, R2, Pages), Docker, Git/GitHub
Web & Platform	React, Node.js, REST APIs, GraphQL
Quantum Toolkits	Qiskit, Cirq (familiarity); custom simulation frameworks for UQC/QRC
Typesetting	LaTeX, Pandoc, Markdown, XeLaTeX, Springer Nature templates
Project Management	Agile/Scrum, Jira, Confluence, MS Project

Certifications

Certification	Issuing Body	Year	Relevance
Project Management Professional (PMP)	PMI	2016	Project leadership, stakeholder management, risk
Contracting Officer's Representative (COR)	DAU	2011	Federal procurement authority, compliance, oversight
American Institute of Certified Planners (AICP)	APA	2009	Policy analysis, data standards, systems planning

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Domain Knowledge

Domain	Depth	Context
Quantum information theory	Expert	p-adic frameworks, adelic rate-distortion, measurement theory
Number theory (p-adic)	Advanced	Foundation for UQC, PANNs, and adelic architectures
Algebraic topology / sheaf theory	Advanced	Measurement stratigraphy, ultrametric geometry

Domain	Depth	Context
Information theory	Expert	*The Information Spectrum* (book), adelic information measures
Signal processing	Advanced	*Alpha Pi Project*, cardiac → quantum signal translation
Transportation modeling	Expert	FHWA Tour-Based Model System, multimodal behavior
Housing & community development	Expert	AARP Livability Index, policy briefs
Legal systems (U.S.)	Proficient	Pro se litigation experience, Empowering Change platform
Federal procurement	Expert	COR certification, $1.5M+ managed

Evidence for all claims is publicly verifiable through published works on Zenodo and ResearchGate.

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